

Vibration Excessive

Symptoms

- Vibration excessive
- Cab noise due to vibration

How To Use This Tree

This symptom tree can be used to troubleshoot all vibration-based symptoms listed above. Start by performing Step 1 troubleshooting. Step 2 will ask a series of questions and will provide a list of troubleshooting steps to perform, depending on the symptoms. Perform the list of troubleshooting steps in the sequence shown in the Specifications/Repair section of the tree.

Shoptalk

Vibration Troubleshooting Documentation Information Questions

- 1) What is the original equipment manufacturer (OEM) make/model?
- 2) What are the mileage/hours?
- 3) Has there been any recent repair and/or maintenance history?
 - Any engine/clutch/transmission rebuild, removal, and installation?
 - History of repeatedly broken brackets and/or capscrews (alternator, fan, exhaust, etc.)?
 - Structural modifications to the vehicle from OEM built?
- 4) Description of vibration?
 - What is shaking (mirror, seat, steering wheel, cab/dash, etc.)?
 - Is there excessive noise in the cab during the vibration?
- 5) What are the conditions when the complaint occurs?
 - Power takeoff (PTO)?
 - Power output (hard pull, during lug down, etc.)?
 - During acceleration and/or deceleration?
 - At idle?
 - With or without a trailer?
 - Does vibration increase with engine speed?
 - Does vibration increase with road speed?
 - Is the vibration at a certain engine rpm?

6) Has the vibration been present since new? (From new, recent repair, modification to equipment)

7) Can the vibration be easily duplicated?

8) Do you have another piece of equipment with the same specification which exhibits the same complaint? (If yes, get the engine serial number (ESN) and possibly test the vehicle)

9) Are you the **only** operator of the equipment? (If no, are the symptoms noticed by other operators)?

General Information

Vibration complaints can be very difficult to troubleshoot and understand the root cause. This troubleshooting document was designed to help guide you through the logical steps of identifying the source. Vibration acceptance is very subjective; what is objectionable to one person can possibly be acceptable to another.

Vibration complaints can be caused by many parts in the system (system includes the engine, driven component, mounts, and equipment). The cause can be transmitted or generated from a remote point that is **not** readily apparent.

Cummins Inc. experience has shown that the engine is rarely the cause of an operator complaint. The majority of the time, it is the engine mounts or design of the various components on the equipment. The engine is **only** at fault if there is a misfire or an engine component that is out of balance.

Vibration complaints that occur **only** at idle speed are most likely caused by the engine mounts. If the engine mount natural frequency is close to the engine firing frequency, the engine will cause the mounts to amplify the normal vibration on an engine idling and cause the adjacent components to vibrate excessively.

Natural Frequency

- Natural frequency, as the name implies, is the frequency at which an object wants to naturally vibrate. The frequency is primarily dependent on mass and elasticity.

Types of Vibration

1) Linear

a) Rotating components

b) Torque reaction

- Caused by unbalanced rotating components and cylinder firing impulses.
- Can be felt and observed visibly.
- When excessive, can cause operator discomfort and damage to components.

2) Torsional - twisting stresses

- Cyclic speeding and slowing of rotating components.
- Controlled by flywheel mass and vibration damper.

- Can **NOT** be felt by the operator.
- Can damage gears and splines.

3) Resonant - component excited at natural frequency

- Is actually linear vibration.
- Resonant vibration occurs when a system or component is excited by linear vibration at its natural frequency.
- Vibration will increase in amplitude as the system's natural frequency is approached. Amplitude will decrease as the exciting forces (engine firing frequency) increase in frequency beyond the system's natural frequency.
- Resonant vibration can be many times larger in amplitude than the exciting force.
- Vibration **must** be controlled by design of mounts (engine and cab) and components.

Engine and Cab Mounts

- The mounts **must** be designed to isolate or reduce the transmission of engine and equipment component vibrations.
- For maximum isolation, it is desired that the natural frequency of the mount be as low as possible.
- Good engine mounts will reduce the amount of engine vibration transmitted to the chassis frame by at least 50 percent at idle.
- Hard engine mounts will provide little or no isolation, and can actually magnify the vibration transmitted to the chassis.
- Stiffness (durometer) and size of the isolator, along with the weight of the engine or component applied are the determining factors when designing a mounting system. An isolator that is correct for one engine, can possibly **not** be right for another. Likewise, because of weight differential, a particular isolator designed for the rear of an engine, probably will **not** be ideal for the front.

Warnings and Cautions

CAUTION

Do not allow the engine to overheat during testing. Engine damage will occur.

Troubleshooting Summary

STEPS		SPECIFICATIONS
STEP 1.	Perform the basic troubleshooting procedures.	
	STEP 1A. Document the information questions in the Shop Talk section of this tree.	Documentation completed?
	STEP 1B. Check for active fault codes or high counts of inactive fault codes.	Active fault codes or high counts of inactive fault codes?
	STEP 1C. Perform the basic troubleshooting checks.	All steps verified to be correct?
	STEP 1D. Determine if the engine is running rough.	Engine running rough?
	STEP 1E. Perform a visual inspection of the engine mounts (without removal).	Visible engine mount damage?
	STEP 1F. Check for an engine mounted component contacting the frame or body.	Engine mounted components touching the frame or body?
	STEP 1G. Check for equipment structural modifications.	Any structural modifications to the equipment present?
	STEP 1H. Complaint since new.	Problem been occurring since the equipment was new?
	STEP 1I. Marine application.	Engine installed in a marine application?
	STEP 1J. Power generator application.	Engine installed in a power generator application?
	STEP 1K. Set to Gains.	Adjusting the gains solved the issue?
	STEP 1L. Do an engine RPM sweep.	Vibration present below 1050 rpm?
	STEP 1M. Do an engine RPM sweep.	Vibration present stationary above 1050 rpm?
	STEP 1N. Check the vibration engine speed range.	Vibration speed range greater than 300 to 400 rpm?
STEP 2.	Advance engine troubleshooting procedures.	
	STEP 2A. Check that the accessory load is not excessive for the idle speed setting.	All steps verified to be correct?

STEPS		SPECIFICATIONS
	STEP 2B. Inspect the engine mounts.	All steps verified to be correct?
	STEP 2C. Inspect the fan hub.	Fan hub meets specifications?
	STEP 2D. Check for malfunctioning belt driven accessories.	Vibration goes away with the drive belts removed?
	STEP 2E. Check for a damaged vibration damper.	Vibration damper damaged or out of specification?
	STEP 2F. Check the overhead adjustments.	Overhead adjustments correct?
	STEP 2G. Check for malfunctioning gear driven components.	Vibration goes away?
	STEP 2H. Check the drive train components.	Disengaging the transmission reduces the effect of the vibration?
	STEP 2I. Check for a loose or damaged flywheel or flex plate.	Flywheel meets specifications?
	STEP 2J. Check the flywheel housing for correct alignment.	Flywheel housing meets specifications?
	STEP 2K. Check for internal engine damage.	Internal engine damage?
STEP 3.	Marine applications.	
	STEP 3A. Check the gear ratio and propeller configuration.	Gear ratio and the propeller incorrectly matched to the engine power?
	STEP 3B. Check for the correct engine mounting isolators and for proper installation requirements.	Engine mount isolators correct and installed correctly?
	STEP 3C. Check for damaged engine mounts and isolators.	Engine mounts and isolators in good condition?
	STEP 3D. Check the exhaust system.	Exhaust system deficient?
	STEP 3E. Check the engine driven accessories.	Engine driven accessory malfunctioning?

STEPS		SPECIFICATIONS
	STEP 3F. Check the shaft coupling to gear coupling alignment.	Shaft coupling to gear coupling misaligned?
	STEP 3G. Check the propeller shaft for proper installation.	Propeller shaft installed correctly?
	STEP 3H. Check the propeller shaft for straightness.	Propeller shaft straightness within the OEM specification?
	STEP 3I. Isolate the engine.	Engine vibration persists?
	STEP 3J. Check for strut/cutlass bearing misalignment.	Strut/cutlass bearing misaligned or strut mounting not secure?
	STEP 3K. Check the propeller.	Propeller out of balance or not fitted properly to the shaft?
	STEP 3L. Check the V-angle on the V-strut.	V-angle on the V-strut matches the angle of the blade on the prop?
	STEP 3M. Check the propeller tunnels.	Entry and exit of the propeller tunnel matches with the propeller blades?
	STEP 3N. Check the engine to transmission torsional coupling.	Torsional coupling incorrect or worn?
	STEP 3O. Check the rudder.	Rudder has excessive clearance in the rudder post?
	STEP 3P. Check the engine flywheel housing to cylinder block alignment.	Flywheel housing alignment incorrect?

STEP 1. Perform the basic troubleshooting procedures.

STEP 1A. Document the information questions in the Shop Talk section of this tree.

Conditions :

- None

Action	Specification/Repair	Next Step

Complete the basic troubleshooting questionnaire. Complete the vibration troubleshooting documentation information questions contained in the Shop Talk section of this procedure.	Documentation completed? YES	1B
	Documentation completed? NO Repair : Complete the documentation.	1A

STEP 1B. Check for active fault codes or high counts of inactive fault codes.

Conditions :

- Turn keyswitch ON.
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step
Check the fault codes. • Use INSITE™ electronic service tool to read the fault codes.	Active fault codes or high counts of inactive fault codes? YES Repair : See the corresponding Electronic Control System Troubleshooting and Repair manual for the engine being serviced.	Repair complete
	Active fault codes or high counts of inactive fault codes? NO	1C

STEP 1C. Perform the basic troubleshooting checks.

Conditions :

- As required.

Action	Specification/Repair	Next Step
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Check or verify the following items before continuing. • Battery voltage is low (engine running) • Lubricating oil level is above specification • External fuel leak • Engine idle speed is set too low • Engine idle speed is set too high • Throttle lever or pedal, return spring, or air throttle damaged or improperly adjusted (use INSITE™ electronic service tool for electronic engines) • Air in the fuel • Fuel pressure • Inlet restriction.	All steps verified to be correct? YES	1D
	All steps verified to be correct? NO	Repair complete

STEP 1D. Determine if the engine is running rough.

Conditions :

- Operate engine at idle speed (less than 900 rpm).
- Turn accessories OFF (air conditioning, fan, PTO).
- Operate engine at appropriate temperature (greater than 77°C [170°F]).

Action	Specification/Repair	Next Step
Determine if the engine is running rough at engine idle or misfiring. Refer to the Engine Runs Rough symptom tree in Section TS or the Engine Performance Troubleshooting Tree in Section TT.	Engine running rough? YES Repair : See Engine Runs Rough symptom tree.	Complete Engine Runs Rough troubleshooting tree
	Engine running rough? NO	1E

STEP 1E. Perform a visual inspection of the engine mounts (without removal).

Conditions :

- Do **not** operate engine.
- Install engine mounts.

Action	Specification/Repair	Next Step
Perform a visual inspection of the engine mounts. Look for obvious damage or something shorting against the mounts, preventing isolation. A more detailed inspection will be carried out later in the procedure.	Visible engine mount damage? YES Repair : Repair or replace the engine mounts. Use Procedure 016-010 in Section 16 in the appropriate service manual.	Repair complete
	Visible engine mount damage? NO	1F

STEP 1F. Check for an engine mounted component contacting the frame or body.

Conditions :

- Do **not** operate engine.
- Install engine mounts.

Action	Specification/Repair	Next Step

Check for an engine mounted component touching the frame or body. Inspect the engine and engine mounted components to make sure none of them are touching the frame and/or body. Including but not limited to the following: • Clamps • Mounting hardware • Exhaust system • Air intake piping • Cooling package support.	Engine mounted components touching the frame or body? YES Repair : Correct the mounting of the engine mounted component.	Repair complete
	Engine mounted components touching the frame or body? NO	1G

STEP 1G. Check for equipment structural modifications.

Conditions : Perform an inspection.		
Action	Specification/Repair	Next Step
Check for any structural modifications to the equipment. Check for any structural modifications to the equipment in the engine area that were completed by the OEM after equipment manufacture. Structural modifications can change the natural frequency of the frame and engine mounting system, which can result in a vibration complaint.	Any structural modifications to the equipment present? YES Repair : Contact the OEM. If possible, remove or isolate the structural modification.	Repair complete
	Any structural modifications to the equipment present? NO	1H

STEP 1H. Complaint since new.

Conditions :

- Record the odometer/hour meter.
- Review the troubleshooting documentation information questions.

Action	Specification/Repair	Next Step
Check the equipment. Check the equipment mileage/hours and compare to the vibration customer interview form completed in Step 1A. • Low mileage/hours is an indication that the complaint has been present since the equipment was new. • Complaints on new equipment are typically due to a manufacturing defect in the system or an inadequate engine mounting design.	Problem been occurring since the equipment was new? YES Repair : The engine mounts are not the right specification for the application, or a structural resonance exists.	Contact a Cummins® Technical Support Specialist or the OEM.
	Problem been occurring since the equipment was new? NO	1I

STEP 1I. Marine application.**Conditions :**

None

Action	Specification/Repair	Next Step
Check if the engine is in a marine application. N/A	Engine installed in a marine application? YES	3A
	Engine installed in a marine application? NO	1J

STEP 1J. Power generator application.**Conditions :**

None

Action	Specification/Repair	Next Step
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Check if the engine is in a power generator application.	Engine installed in a power generator application? YES	1K
	Engine installed in a power generator application? NO	1L

STEP 1K. Set to gains.

Conditions :

Operate engine under load.

Action	Specification/Repair	Next Step
Adjust the gains to stabilize the engine. If other generator sets of the same type are operating in the same area without issue, determine the gains settings on those units and copy to this unit.	Adjusting the gains solved the issue? YES	Repair complete.
	Adjusting the gains solved the issue? NO	1L

STEP 1L. Do an engine rpm sweep.

Conditions :

- Operate engine.
- Connect INSITE™ electronic service tool.
- Do **not** move vehicle.

Action	Specification/Repair	Next Step
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Perform a slow (at 4 rpm per second) rpm sweep and observe where the vibration occurs. Record the engine speed at which any unusual vibration related noise occurs (mirrors, panels, doors, seat, etc.). Record the speed points or ranges with excessive vibration. If a resonance is passed through quickly and is getting up to the operating speed range, and does not exist in the idle speed or peak operating range, it represents no major problem.	Vibration present below 1050 rpm? YES	1N
	Vibration present below 1050 rpm? NO	1M

STEP 1M. Do an engine rpm sweep.

Conditions :

- Operate engine.
- Connect INSITE™ electronic service tool.
- Do **not** move vehicle.

Action	Specification/Repair	Next Step
Perform a slow (at 4 rpm per second) rpm sweep and observe where the vibration occurs. If the vibration increases progressively from idle to maximum speed, rotating or reciprocating unbalance is the source. This can be caused by any rotating components or engine mount isolation.	Vibration present above 1050 rpm? YES	1L
	Vibration present above 1050 rpm? NO	Repair complete

STEP 1N. Check the vibration engine speed range.

Conditions :

- Operate engine.
- Connect INSITE™ electronic service tool.
- Do **not** move vehicle.

Action	Specification/Repair	Next Step
Perform a slow (at 4 rpm per second) rpm sweep and observe where the vibration occurs. This step is to identify if the vibration progressively increases with engine speed or if it starts and stops within a slow engine rpm band. If the vibration progressively increases with engine speed and has a peak band greater than 300 rpm, this can indicate a rotating component that is out of balance. If the vibration peak is in a tight band of approximately 300 to 400 rpm or less, this indicates that a structural component of the engine or equipment is going into resonance because its natural frequency is close to or the same as the engine firing frequency.	Vibration speed range greater than 300 to 400 rpm? YES	2A
	Vibration speed range greater than 300 to 400 rpm? NO	2A

STEP 2. Advanced engine troubleshooting procedures.**STEP 2A. Check that the accessory load is not excessive for the idle speed setting.**

Conditions :

- Operate engine at idle speed (less than 900 rpm).
- Turn OFF accessories (air conditioning, fan, and PTO).
- Operate engine at operating temperature (greater than 77°C [170°F]).

Action	Specification/Repair	Next Step
Disable all engine driven accessories and PTOs to make sure they are not applying excessive load to the engine. N/A	All steps verified to be correct? YES	2B
	All steps verified to be correct? NO Repair : Repair as required.	Repair complete

STEP 2B. Inspect the engine mounts.**Conditions :**

- Do **not** operate engine.
- Remove engine mount isolators.
- Install engine mount isolators.

Action	Specification/Repair	Next Step
Inspect the engine mount brackets, isolators, and mounting hardware. • Check the engine mount isolators for installation damage. • Check the alignment of the engine mount brackets. • Check for premature wear on the engine mount isolators and mounting hardware.	All steps verified to be correct? YES	2C
	All steps verified to be correct? NO Repair : Repair or replace the damaged components. Use Procedure 016-010 in Section 16 or the appropriate service manual.	Repair complete

STEP 2C. Inspect the fan hub.

Conditions :

- Do **not** operate engine.

Action	Specification/Repair	Next Step
Inspect the fan hub to see if it is loose, damaged, or has excessive hub bearing end play. Use Procedure 008-040 in the appropriate service manual for fan hub specifications.	Fan hub meets specifications? YES	2D
	Fan hub meets specifications? NO Repair : Repair or replace the fan hub as required. Use Procedure 008-040 in the appropriate service manual.	Repair complete

STEP 2D. Check for malfunctioning belt driven accessories.**Conditions :**

- Remove the drive belts.

Action	Specification/Repair	Next Step
Check the belt driven accessories. Remove the drive belts and operate the engine under the condition where vibration occurs.	Vibration goes away with the drive belts removed? YES Repair : Repair or replace the malfunctioning belt driven accessory. See equipment manufacturer service information.	2D
	Vibration goes away with the drive belts removed? NO	2E

STEP 2E. Check for a damaged vibration damper.

Conditions :

- Do **not** operate engine.

Action	Specification/Repair	Next Step
Remove and visually inspect the vibration damper. Use Procedure 001-052 in the appropriate service manual for vibration damper inspection specifications.	Vibration damper damaged or out of specification? YES Repair : Replace the vibration damper. Use Procedure 001-052 in the appropriate service manual for replacement instructions.	Repair complete
	Vibration damper damaged or out of specification? NO	2F

STEP 2F. Check the overhead adjustments.**Conditions :**

- Do **not** operate engine.
- Remove rocker lever cover.

Action	Specification/Repair	Next Step
Measure and adjust the overhead settings. • Check the overhead components for damages. Use Procedure 003-004 in the appropriate service manual.	Overhead adjustments correct? YES	2G
	Overhead adjustments correct? NO Repair : Repair or adjust the overhead. Use Procedure 003-004 in the appropriate service manual.	Repair complete

STEP 2G. Check for malfunctioning gear driven accessories.

Conditions : None		
Action	Specification/Repair	Next Step
Check the gear driven accessories for any damage or signs of wear. If possible, isolate any gear-driven accessories and check for vibration.	Vibration goes away? YES Repair : Repair or replace the damaged gear driven component. Use the appropriate procedure in the appropriate service manual.	Repair complete
	Vibration goes away? NO	2H

STEP 2H. Check the drive train components.

Conditions : - Isolate the engine from the transmission. - Operate engine.		
Action	Specification/Repair	Next Step
With the transmission disengaged from the engine, operate the engine in the condition where the vibration occurs. If there is a significant vibration reduction, the transmission is the source of the vibration.	Disengaging the transmission reduces the effect of the vibration? YES Repair : Repair or replace the transmission. See equipment manufacturer service information.	Repair complete
	Disengaging the transmission reduces the effect of the vibration? NO	2I

STEP 2I. Check for a loose or damaged flywheel or flex plate.

Conditions : Remove transmission.		
Action	Specification/Repair	Next Step
Check the flywheel. - Check the flywheel bore and face runout. - Check the flywheel for damage. Use Procedure 016-005 in the appropriate service manual.	Flywheel meets specifications? YES	2J
	Flywheel meets specifications? NO Repair : Repair or replace the flywheel or flexplate. Use Procedure 016-005 in the appropriate service manual.	Repair complete

STEP 2J. Check the flywheel housing for correct alignment.

Conditions : - Transmission removed. - Flywheel/flexplate removed.		
Action	Specification/Repair	Next Step
Check the flywheel housing bore and face alignment. Use Procedure 016-006 in the appropriate service manual.	Flywheel housing meets specifications? YES	2K
	Flywheel housing meets specifications? NO Repair : Repair or replace the flywheel housing. Use Procedure 016-006 in the appropriate service manual.	Repair complete

STEP 2K. Check for internal engine damage.

Conditions : • None		
Action	Specification/Repair	Next Step
Contact a support specialist. At this point, a significant amount of labor has been invested in the repair. Before disassembly the engine, seek troubleshooting assistance. Contact the appropriate Technical Support Channel for the facility. They will provide the necessary guidance and schedule on-site support, if deemed necessary. • Camshaft journals and number 1 camshaft busing are severely damaged. • Gear train backlash is excessive or the gear teeth are damaged. • Idler gear bushing damaged or worn. • Main or connecting rod bearings damaged. • Gears out of balance or gear bushing damage. • Connecting rod damage.	Internal engine damage? YES	Contact Cummins® Technical Support
	Internal engine damage? NO	Contact Cummins® Technical Support

STEP 3. Marine applications.

STEP 3A. Check the gear ratio and propeller configuration.

Conditions :

- Turn keyswitch OFF.

Action	Specification/Repair	Next Step
Check for an incorrect matching of the gear ratio and propeller to the engine power.	Gear ratio and the propeller incorrectly matched to the engine power? YES Repair : Contact a Cummins® Distributor or a Marine District Field Service Manager.	Repair complete
	Gear ratio and the propeller incorrectly matched to the engine power? NO	3B

STEP 3B. Check for the correct engine mounting isolators and for proper installation requirements.**Conditions :**

- None.

Action	Specification/Repair	Next Step
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Check for the correct engine mount isolators and for propeller installation requirements.	Engine mount isolators correct and installed correctly? YES	3C
	Engine mount isolators correct and installed correctly? NO Repair : Check for proper isolator installation requirements. Replace and repair vibration isolators as needed. Use Procedure 016-026 in the appropriate service manual and Engine Mounting/Drive Systems section in the Marine Commercial Installation Directions, Bulletin 3884744. If the isolators are not manufactured by Cummins Inc., see equipment manufacturer service information.	Repair complete

STEP 3C. Check for damaged engine mounts and isolators.

Conditions : • None.		
Action	Specification/Repair	Next Step

Inspect the engine mount and isolators for damage.	Engine mounts and isolators in good condition? YES	3D
	Engine mounts and isolators in good condition? NO Repair : Remove and replace the engine mount isolators. Use Procedure 016-026 in the appropriate service manual and Engine Mounting/Drive Systems section in the Marine Commercial Installation Directions, Bulletin 3884744. If the isolators are not manufactured by Cummins Inc., see equipment manufacturer service information.	Repair complete

STEP 3D. Check the exhaust system.

Conditions :

- None.

Action	Specification/Repair	Next Step
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Check for exhaust system deficiencies.	Exhaust system deficient? YES Repair : Repair or replace as needed. See the Exhaust System section in the Marine Commercial Installation Directions, Bulletin 3884744, and equipment manufacturer service information.	Repair complete
	Exhaust system deficient? NO	3E

STEP 3E. Check the engine driven accessories.

Conditions : • Turn keyswitch ON. • Turn keyswitch OFF.		
Action	Specification/Repair	Next Step
Check for engine driven accessory malfunctions. • Isolate or disconnect the accessories and check for vibration. • Do not operate the engine if the sea water pump is disconnected.	Engine driven accessory malfunctioning? YES Repair : Determine the cause of the malfunctioning accessories and correct the problem. See the Exhaust System section in the Marine Commercial Installation Directions, Bulletin 3884744, and equipment manufacturer service information.	Repair complete
	Engine driven accessory malfunctioning? NO	3F

STEP 3F. Check the shaft coupling to gear coupling alignment.**Conditions :**

- Turn keyswitch OFF.

Action	Specification/Repair	Next Step
Check the shaft coupling to gear coupling alignment.	Shaft coupling to gear coupling misaligned? YES Repair : Repair or replace as needed. Use Procedure 016-025 in the appropriate service manual and the Engine Mounting/Drive Systems section in the Marine Commercial Installation Directions, Bulletin 3884744, and gear manufacturer recommendations.	Repair complete
	Shaft coupling to gear coupling misaligned? NO	3G

STEP 3G. Check the propeller shaft for proper installation.**Conditions :**

- None.

Action	Specification/Repair	Next Step
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Check the propeller shaft for proper installation.	Propeller shaft installed correctly? YES	3H
	Propeller shaft installed correctly? NO Repair : Repair or replace as needed. Use Procedure 016-025 in the appropriate service manual and the Engine Mounting/Drive Systems section in the Marine Commercial Installation Directions, Bulletin 3884744, and the gear manufacturer's recommendations.	Repair complete

STEP 3H. Check the propeller shaft for straightness.

Conditions : • None.		
Action	Specification/Repair	Next Step
Check the propeller shaft for straightness.	Propeller shaft straightness within the OEM specification? YES	3I
	Propeller shaft straightness within the OEM specification? NO Repair : Repair or replace the propeller shaft as needed. See an Authorized OEM Service Location.	Repair complete

STEP 3I. Isolate the engine.

Conditions :

- Disconnect the drive shaft.

Action	Specification/Repair	Next Step
Run the engine without the drive shaft attached at the coupler.	Engine vibration persists? YES Repair : Check the engine vibration damper for damage. Repair or replace as needed. Use Procedure 001-052 in the appropriate service manual.	Repair complete
	Engine vibration persists? NO	3J

STEP 3J. Check for strut/cutlass bearing misalignment.**Conditions :**

- Turn keyswitch OFF.

Action	Specification/Repair	Next Step
Check for strut/cutlass bearing wear or damage. Check for strut/cutlass bearing misaligned or strut mounting is not secure.	Strut/cutlass bearing damaged, misaligned, or strut mounting not secure? YES Repair : Check the strut for mounting stiffness. Repair or replace as necessary. See an Authorized OEM Service Location.	Repair complete
	Strut/cutlass bearing damaged, misaligned, or strut mounting not secure? NO	3K

STEP 3K. Propeller checks.

Conditions : None.		
Action	Specification/Repair	Next Step
Check for propeller out-of-balance or propeller not fitted properly to shaft. Check for any propeller damage. Check for excessive propeller cavitation. See equipment manufacturer service information for propeller cavitation specifications.	Propeller out of balance or not fitted properly to the shaft? YES	3L
	Propeller out of balance or not fitted properly to the shaft? NO Repair : Repair or replace the propeller. See equipment manufacturer service information.	Repair complete

STEP 3L. Check the V-angle on the V-strut.

Conditions : None.		
Action	Specification/Repair	Next Step
Check to see if the V-angle on the V-strut does not match the angle of the blade on the propeller.	V-angle on the V-strut matches the angle of the blade on the prop? YES	3M
	V-angle on the V-strut matches the angle of the blade on the prop? NO Repair : Contact the OEM as a possible re-design of the V-angle on the V-strut may be required.	Repair complete

STEP 3M. Check the propeller tunnels.

Conditions : None.		
Action	Specification/Repair	Next Step
Check if the propeller tunnels are properly matched with the propellers.	Entry and exit of the propeller tunnel matches with the propeller blades? YES	3N
	Entry and exit of the propeller tunnel matches with the propeller blades? NO Repair : Repair or replace as needed. See an Authorized OEM Service Location.	Repair complete

STEP 3N. Check the engine to transmission torsional coupling.

Conditions : None.		
Action	Specification/Repair	Next Step
Check the engine to transmission torsional coupling.	Torsional coupling incorrect or worn? YES Repair : Replace the coupling. See an Authorized OEM Service Location.	Repair complete
	Torsional coupling incorrect or worn? NO	30

STEP 30. Check the rudder.

Conditions : None.		
Action	Specification/Repair	Next Step

Check the rudder for excessive play in the rudder post.	Rudder has excessive clearance in the rudder post? YES Repair : Repair or replace as needed. See an Authorized OEM Service Location.	Repair complete
	Rudder has excessive clearance in the rudder post? NO	3P

STEP 3P. Check the engine flywheel housing to cylinder block alignment.

Conditions : None.		
Action	Specification/Repair	Next Step

Check the engine flywheel housing to cylinder block alignment.	Flywheel housing alignment incorrect? YES Repair : Align the flywheel housing to cylinder block. Use Procedure 016-006 in the appropriate service manual.	Repair complete
	Flywheel housing alignment incorrect? NO Repair : The engine can possibly have internal damage that has not been detected. Analyze the oil and inspect the filters to locate an area of probable damage. Use Procedure 007-083 in the appropriate service manual. The engine can possibly need to be rebuilt. Use Procedure 000-001 and the engine rebuild specifications in the appropriate service manual. If the engine is not damaged, the problem can possibly be the vessel design. See an Authorized OEM Service Location.	Repair complete